

CS 5180 – Project Report

Maximum Points: 100 pts (80% of the project*).

[1 point] Cover page (see example below)

Project Title
Team Members

Cover Page Example

[Project Title]

Team Members

Student 1	Student 2	Student 3	Student 4
[Name of Student]	[Name of Student]	[Name of Student]	[Name of Student]

[2 point] List of Content (new page)

[1 point] List of Figures (new page)

[1 point] List of Tables (new page)

List of Content Example

List OF FIGURES	1
LIST OF TABLES	2
PROJECT PLAN	3
1 INTRODUCTION	3
2 ORGANIZATION OF THE PROJECT	4
3 METHODS AND TECHNIQUES	
4 ...	
REFERENCES	20

* The other 20% of the project grade will be assigned to the project presentation.

[9 points] Project Plan

(new page)

Introduction

Brief introduction of the project describing its goal.

Team roles and responsibilities

Specification of each team member's role and responsibilities.

Data, methods, and techniques

High-level overview of the data, methods, and techniques chosen for system design and implementation, including dataset, text processing, indexing, retrieval model(s), and evaluation metrics used.

[8 points] Dataset Description and Analysis

(new page)

Dataset statistics

Information about number of documents, queries, documents and queries average length, and relevant documents per query.

[9 points] Text Processing

(new page)

Tokenizer definition

Description of how documents have been tokenized to produced index terms (including phrases, if applicable). Provide application example(s).

Surface level normalization definition

Description of how tokens have been normalized at surface level (e.g., lowercasing, handling of punctuation, numbers, or special symbols). Provide application example(s).

Stopping definition

Description of how stopping words have been handled. Provide application example(s).

Stemming/Lemmatizing definition

Description of how stemming or lemmatizing has been conducted. Provide application example(s).

[8 points] Index Construction

(new page)

Term weighting definition

Description of the statistical method used to weight index terms (TF-IDF, BIM, BM25, or Language Model components). Provide application example(s).

Index design

Description of how the inverted index has been constructed, including the construction method (e.g., hash-based or sort-based), the design variant (e.g., binary, tf-based, positional, score-based), and any additional data stored (e.g., document frequency or document length). Provide application example(s).

[6 points] Retrieval Model

(new page)

Ranking

Information about the statistical IR model used for ranking (e.g. VSM, BIM, BM25, or Language Model), including the scoring function, how documents and query (if applicable) are represented, how relevance scores are computed, and how documents are ranked. Provide application example(s).

[50 points] Implementation

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Source code:

A GitHub repository containing the complete Python implementation of your IR system.

Video demonstration:

A minimum 5-minute and maximum 10-minute video containing an overall system explanation with demonstration of the query processing and display of ranked results.

[4 points] Discussion

(new page)

Analysis of the results obtained and lessons learned.

[1 point] References